

Physics - Question 81

Original paper page 23; original question and option regions

81. The pressure on a circular plate is measured by measuring the force on the plate and the radius of the plate. If the errors in measurement of the force and the radius are 5% and 3% respectively, the percentage of error in the measurement of pressure is

1) 8

2) 14

3) 11

4) 12

Sol:
$$\frac{\Delta p}{p} \times 100 = \left(\frac{\Delta F}{F} + \frac{2\Delta r}{r} \right) 100 = 5 + 2 \times 3 = 11$$

Physics - Question 82

Original paper page 23; original question and option regions

82. A body is projected vertically from the surface of the earth of radius 'R' with a velocity equal to half of the escape velocity. The maximum height reached by the body is

1) $R/2$

2) $R/3$

3) $R/4$

4) $R/5$

Sol:
$$-\frac{GMm}{R} + \frac{1}{2} m \frac{V_e^2}{4} = \frac{-GMm}{R+h}$$

$$\frac{1}{2} m \frac{V_e^2}{4} = + \frac{GMm}{R} - \frac{GMm}{R+h}$$

$$\frac{2GM}{8R} = \frac{GMh}{R(R+h)}$$

$$\frac{1}{4} = \frac{h}{R+h}$$

$$R+h = 4h$$

$$h = R/3$$

Physics - Question 83

Original paper page 23; original question and option regions

83. A particle aimed at a target, projected with an angle 15° with the horizontal is short of the target by 10m. If projected with an angle of 45° is away from the target by 15m, then the angle of projection to hit the target is

- 1) $\frac{1}{2} \sin^{-1} \left(\frac{1}{10} \right)$ 2) $\frac{1}{2} \sin^{-1} \left(\frac{3}{10} \right)$ 3) $\frac{1}{2} \sin^{-1} \left(\frac{9}{10} \right)$ 4) $\frac{1}{2} \sin^{-1} \left(\frac{7}{10} \right)$

Sol: $R - 10 = \frac{u^2 \sin 30}{g} \Rightarrow R + 10 = \frac{u^2}{g}$

$$\frac{R - 10}{R + 10} = \frac{1}{2} \Rightarrow R = 35$$

$$\frac{u^2}{g} = 50$$

$$\theta = \frac{1}{2} \sin^{-1} (7/10)$$

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Physics - Question 83 (continued)

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Physics - Question 84

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84. A man running at a speed of 5kmph finds that the rain falls vertically. When he stops running he finds that the rain is falling at an angle of 60° with the horizontal. The velocity of rain with respect to running man is

- 1) $\frac{5}{\sqrt{3}}$ kmph 2) $\frac{5\sqrt{3}}{2}$ kmph 3) $\frac{4\sqrt{3}}{5}$ kmph 4) $5\sqrt{3}$ kmph

Sol: $V_R = x\mathbf{i} - y\mathbf{j}$

$$V_m = 5\mathbf{i}$$

$$V_{Rm} = (x - 5)\mathbf{i} - y\mathbf{j}$$

$$X = 5$$

$$y/x = \sqrt{3} = \tan 60^\circ$$

$$y = 5\sqrt{3}$$

Physics - Question 85

Original paper page 24; original question and option regions

85. A horizontal force just sufficient to move a body of mass 4kg lying on a rough horizontal surface, is applied on it. Coefficient of static and kinetic frictions are 0.8 and 0.6 respectively. If the force continues to act even after the body has started moving, the acceleration of the body is ($g = 10\text{ms}^{-2}$)

1) 6ms^{-2}

2) 8ms^{-2}

3) 2ms^{-2}

4) 4ms^{-2}

Sol: $a = (\mu_s - \mu_k) g$

$$= (0.8 - 0.6) 10$$

$$= 2 \text{ m/s}^2$$

Physics - Question 86

Original paper page 24; original question and option regions

86. A force $(2\hat{i} + \hat{j} - k)\text{N}$ acts on a body which is initially at rest. At the end of 20 sec the velocity of the body is $(4\hat{i} + 2\hat{j} - 2k)\text{ms}^{-1}$, then the mass of the body is

1) 8 kg

2) 10 kg

3) 5 kg

4) 4.5 kg

Sol: $V = \frac{F}{m}t$

$$\sqrt{24} = \frac{\sqrt{6} \times 20}{m}$$

$$M = 10 \text{ kg}$$

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Physics - Question 86 (continued)

Original paper page 25; original question and option regions

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Physics - Question 87

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87. A man of weight 50kg carries an object to a height of 20m in a time of 10sec. The power used by the man in this process is 2000W, then find the weight of the object carried by the man [assume $g = 10 \text{ ms}^{-2}$]

1) 100 kg

2) 25 kg

3) 50 kg

4) 10 kg

Sol:
$$p = \frac{(M + m)gh}{t}$$

$$2000 = \frac{(50 + m)10 \times 20}{10}$$

$$m = 50 \text{ kg}$$

Physics - Question 88

Original paper page 25; original question and option regions

- 88. A ball 'P' moving with a speed of $v \text{ ms}^{-1}$ collides directly with another identical ball 'Q' moving with a speed 10 ms^{-1} in the opposite direction. P comes to rest after the collision. If the coefficient of restitution is 0.6, the value of v is**

- 1) 30 ms^{-1} 2) 40 ms^{-1} 3) 50 ms^{-1} 4) 60 ms^{-1}



Sol:

$$V_1 = \frac{m_1 - em_2}{m_1 + m_2} u_1 + \frac{(1+e)m_2}{m_1 + m_2} u_2 = \frac{1-e}{2} v + \frac{1+e}{2} (-10)$$

$$0 = \frac{1-0.6}{2} v + \frac{1.6}{2} (-10)$$

$$0 = \frac{0.4}{2} v - 8$$

$$0.2v = 8$$

$$v = 40$$

Physics - Question 89

Original paper page 25; original question and option regions

89. A particle of mass $m = 5$ units is moving with uniform speed $V = 3\sqrt{2}$ units in the XY plane along the line $Y = X + 4$. The magnitude of the angular momentum about origin is

1) zero

2) 60 units

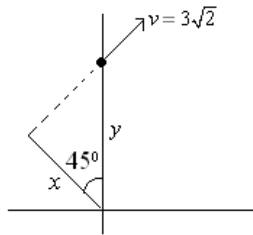
3) 7.5 units

4) 40 units

Sol: $L = mvr \sin \theta$

$$= 5 \times 3\sqrt{2} \times 4 \times \frac{1}{\sqrt{2}}$$

$$= 60$$



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Physics - Question 89 (continued)

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Physics - Question 90

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90. The kinetic energy of a circular disc rotating with a speed of 60 r.p.m. about an axis passing through a point on its circumference and perpendicular to its plane is (mass of circular disc = 5kg, radius of disc = 1 m) approximately.

1) 170 J

2) 160 J

3) 150 J

4) 140 J

Sol: $\frac{1}{2} \left[\frac{3}{2} MR^2 \right] \left(60 \times \frac{2\pi}{60} \right)^2$

$$\frac{3}{4} 5(1)^2 \times 4\pi^2$$

$$15\pi^2$$

$$\approx 150\text{J}$$

Physics - Question 91

Original paper page 26; original question and option regions

91. The amplitude of a simple pendulum is 10cm. When the pendulum is at a displacement of 4cm from the mean position, the ratio of kinetic and potential energies at the point is

1) 5.25

2) 2.5

3) 4.5

4) 7.5

Sol:
$$\frac{KE}{PE} = \frac{A^2 - x^2}{x^2} = \frac{100 - 16}{16} = \frac{84}{16}$$
$$= 5.25$$

Physics - Question 92

Original paper page 26; original question and option regions

92. A satellite revolving around a planet has orbital velocity 10 km/s. The additional velocity required for the satellite to escape from the gravitational field of the planet is

1) 14.14 km/s

2) 11.2 km/s

3) 4.14 km/s

4) 41.4 km/s

Sol: $\Delta v = v_e - v_o$

$$= (\sqrt{2} - 1)v_o$$

$$= 0.414 v_o$$

$$= 0.414 \times 10$$

$$= 4.14$$

Physics - Question 93

Original paper page 26; original question and option regions

93. The length of a metal wire is l_1 when the tension in it is F_1 and l_2 when the tension is F_2 .

Then original length of the wire is

1) $\frac{l_1 F_1 + l_2 F_2}{F_1 + F_2}$

2) $\frac{l_2 - l_1}{F_2 - F_1}$

3) $\frac{l_1 F_2 - l_2 F_1}{F_2 - F_1}$

4) $\frac{l_1 F_1 - l_2 F_2}{F_2 - F_1}$

Sol: $l_o = \frac{l_1 T_2 - l_2 T_1}{T_2 - T_1}$

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Physics - Question 93 (continued)

Original paper page 27; original question and option regions

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Physics - Question 94

Original paper page 27; original question and option regions

94. The average depth of Indian ocean is about 30000m. The value of fractional compression $\left(\frac{\Delta V}{V}\right)$ of water at the bottom of the ocean is (given that the bulk modulus of water is $2.2 \times 10^9 \text{ Nm}^{-2}$, $g = 9.8 \text{ ms}^{-2}$, $\rho_{\text{H}_2\text{O}} = 1000 \text{ kg.m}^{-3}$)

1) 3.4×10^{-2}

2) 1.34×10^{-2}

3) 4.13×10^{-2}

4) 13.4×10^{-2}

Sol:
$$\frac{\Delta V}{V} = \frac{p}{k} = \frac{h \rho g}{k} = \frac{3 \times 10^3 \times 10^3 \times 9.8}{2.2 \times 10^9}$$

$$\approx \frac{29.4}{2.2} \times 10^{-3}$$

$$\approx \frac{2.9}{2.2} \times 10^{-2}$$

$$\approx 1.34 \times 10^{-2}$$

Physics - Question 95

Original paper page 27; original question and option regions

95. The ratio of energies of emitted radiation by a black body at 600 k and 900 k when the surrounding temperature is 300 k

1) $\frac{5}{16}$

2) $\frac{7}{16}$

3) $\frac{3}{16}$

4) $\frac{9}{16}$

Sol: $\frac{Q}{At} \propto (T^4 - T_s^4)$

$$\frac{Q_1}{Q_2} = \frac{600^4 - 300^4}{900^4 - 300^4} = \frac{2^4 - 1}{3^4 - 1} = \frac{15}{18} = \frac{5}{6}$$

Physics - Question 96

Original paper page 27; original question and option regions

96. The specific heat of helium at constant volume is $12.6 \text{ Jmol}^{-1}\text{K}^{-1}$. The specific heat of helium at constant pressure in $\text{Jmol}^{-1}\text{K}^{-1}$ is about (Assume the temperature of the gas is moderate, universal gas constant, $R = 8.314 \text{ Jmol}^{-1}\text{K}^{-1}$)

1) 12.6

2) 16.8

3) 18.9

4) 21

Sol: $c_p = C_v + R = 12.6 + 8.3$

≈ 20.9

≈ 21

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Physics - Question 96 (continued)

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Physics - Question 97

Original paper page 28; original question and option regions

97. A gas does 4.5 J of external work during adiabatic expansion. If its temperature falls by 2K, then its internal energy will be

1) increased by 4.5 J

2) decreased by 4.5 J

3) decreased by 2.25 J

4) increased by 9.0 J

Sol: $dQ = dU + dW$

$$0 = dU + dW$$

$$dU = -dW = -4.5 \text{ J}$$

Physics - Question 98

Original paper page 28; original question and option regions

98. The relation between efficiency ‘ η ’ of a heat engine and the co-efficient of performance ‘ α ’ of a refrigerator is

1) $\eta = \frac{1}{1-\alpha}$

2) $\eta = \frac{1}{1+\alpha}$

3) $\eta = 1+\alpha$

4) $\eta = 1-\alpha$

Sol:
$$\eta = \frac{1}{1+\beta} = \frac{1}{1+\frac{T_2}{T_1-T_2}} = \frac{T_1-T_2}{T_1}$$

$$= 1 - \frac{T_2}{T_1}$$

Physics - Question 99

Original paper page 28; original question and option regions

99. A flask contains argon and chlorine in the ratio of 2 : 1 by mass. The temperature of the mixture is 27°C . The ratio of average kinetic energies of two gases per molecule is

1) 1 : 1

2) 2 : 1

3) 3 : 1

4) 6 : 1

Sol: K.E per molecule $\frac{3}{2}KT$ so 1 : 1

Physics - Question 100

Original paper page 28; original question and option regions

100. A transverse wave is represented by the equation $y = 2\sin (30t - 40x)$ and the measurements of distances are in meters, then the velocity of propagation is

1) 15ms^{-1}

2) 0.75ms^{-1}

3) 3.75ms^{-1}

4) 30ms^{-1}

Sol: $v = \frac{w}{k} = \frac{30}{40} = 0.75\text{m/s}$

Physics - Question 101

Original paper page 28; original question and option regions

101. Two closed pipes have the same fundamental frequency. One is filled with oxygen and the other with hydrogen at the same temperature. Ratio of their lengths respectively is

1) 1 : 4

2) 4 : 1

3) 1 : 2

4) 2 : 1

Sol:
$$\frac{v_0}{4l_0} = \frac{v_H}{4l_H} = \frac{l_0}{l_H} = \frac{v_0}{v_H} = \sqrt{\frac{m_H}{M_0}} = \sqrt{\frac{2}{32}} = \frac{1}{4}$$

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Physics - Question 101 (continued)

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Physics - Question 102

Original paper page 29; original question and option regions

102. An image is formed at a distance of 100cm from the glass surface when light from point source in air falls on a spherical glass surface with refractive index 1.5. The distance of the light source from the glass surface is 100 cm. The radius of curvature is

1) 20 cm

2) 40 cm

3) 30 cm

4) 50 cm

Sol:
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{100} + \frac{1}{100} = \frac{0.5}{R}$$

$$\frac{1}{100} \times 2.5 = \frac{0.5}{R} \Rightarrow R = 20\text{cm}$$

Physics - Question 103

Original paper page 29; original question and option regions

103. Two coherent sources of intensity ratio 9 : 4 produce interference. The intensity ratio of maxima and minima of the interference pattern is

1) 13 : 5

2) 5 : 1

3) 25 : 1

4) 3 : 2

Sol:
$$\frac{I_{\text{Max}}}{I_{\text{Min}}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{\left(\sqrt{\frac{I_1}{I_2}} + 1\right)^2}{\left(\sqrt{\frac{I_1}{I_2}} - 1\right)^2} = \frac{\left(\frac{3}{2} + 1\right)^2}{\left(\frac{3}{2} - 1\right)^2} = \frac{\frac{25}{4}}{\frac{1}{4}} = 25:1$$

Physics - Question 104

Original paper page 29; original question and option regions

104. The energy of a parallel plate capacitor when connected to a battery is E. With the battery still in connection, if the plates of the capacitor are separated so that distance between them is twice the original distance, then the electrostatic energy becomes

1) $2E$

2) $E/4$

3) $E/2$

4) $4E$

Sol: $E = \frac{1}{2}CV^2 = \frac{1}{2} \frac{\epsilon_0 A}{d} V^2$

$$E^1 = \frac{1}{2} \frac{\epsilon_0 A}{2d} V^2 = \frac{E}{2}$$

Physics - Question 105

Original paper page 29; original question and option regions

105. Two point charges $+8\mu\text{C}$ and $+12\mu\text{C}$ repel each other with a force of 48N . When an additional charge of $-10\mu\text{C}$ is given to each of these charges (the distance between the charges is unaltered) then the new force is

1) Repulsive force of 24N 2) Attractive force of 24N 3) Repulsive force of 2N 4) Attractive force of 2N

Sol: $F = \frac{k \times 8 \times 12}{d^2}$ $F^1 = \frac{k \times 2 \times 2}{d^2}$

$$\frac{F^1}{F} = \frac{4}{8 \times 12} \Rightarrow F^1 = \frac{F}{24} = \frac{48}{24} = 2\text{N}$$

Attractive force of 2N

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Physics - Question 105 (continued)

Original paper page 30; original question and option regions



Physics - Question 106

Original paper page 30; original question and option regions

106. If the dielectric constant of substance is $K = \frac{4}{3}$, then the electric susceptibility χ_e is

1) $\frac{\epsilon_0}{3}$

2) $3\epsilon_0$

3) $\frac{4}{3}\epsilon_0$

4) $\frac{3}{4}\epsilon_0$

Sol: $\chi = (K - 1)\epsilon_0 = \frac{\epsilon_0}{3}$

Physics - Question 107

Original paper page 30; original question and option regions

107. In a region of uniform electric field of intensity E , an electron of mass m_e is released from rest. The distance travelled by the electron in a time ' t ' is

1) $\frac{2m_e t^2}{e}$

2) $\frac{eEt^2}{2m_e}$

3) $\frac{m_e g t^2}{eE}$

4) $\frac{2Et^2}{em_e}$

Sol: $S = ut + \frac{1}{2}at^2$

$$S = (0)t + \frac{1}{2} \frac{Ee}{m_e} t^2$$

$$S = \frac{1}{2} \frac{Ee}{m_e} t^2$$

Physics - Question 108

Original paper page 30; original question and option regions

108. A constant potential difference is applied between the ends of the wire. If the length of the wire is elongated 4 times, then the drift velocity of electrons will be

- 1) increases 4 times 2) decreases 4 times 3) increases 2 times 4) decreases 2 times

Sol: $i = nc AV_d \Rightarrow \frac{V}{R} = neAV_d$

$$\frac{V}{\rho \times l} \times A = neAV_d \Rightarrow V_d \propto \frac{1}{l}$$

So drift velocity decreases 4 times

Physics - Question 109

Original paper page 30; original question and option regions

109. In a metre bridge, the gaps are enclosed by resistances of 2Ω and 3Ω . The value of shunt to be added to 3Ω resistor to shift the balancing point by 22.5 cm is

1) 1Ω

2) 2Ω

3) 2.5Ω

4) 5Ω

Sol: $\frac{2}{3} = \frac{l}{100-l}$

$$l = 40\text{cm}$$

$$\frac{\frac{2}{3x}}{3+x} = \frac{40+22.5}{100-62.5} \Rightarrow x = 2\Omega$$

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Physics - Question 109 (continued)

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Physics - Question 110

Original paper page 31; original question and option regions

110. Two long straight parallel conductors 10cm apart, carry equal currents of magnitude 3A in the same direction. Then the magnetic induction at a point midway between them is

1) $2 \times 10^{-5} \text{ T}$

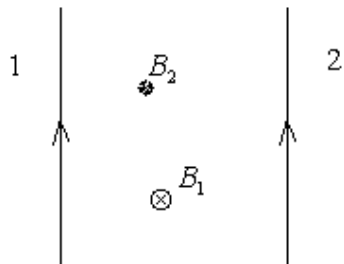
2) $3 \times 10^{-5} \text{ T}$

3) Zero

4) $4 \times 10^{-5} \text{ T}$

Sol: $B = B_1 - B_2$

$$B_1 = B_2 \Rightarrow B = 0$$



Physics - Question 111

Original paper page 31; original question and option regions

111. In a crossed field, the magnetic field induction is 2.0T and electric field intensity is $20 \times 10^3 \text{ V/m}$. At which velocity the electron will travel in a straight line without the effect of electric and magnetic fields?

- 1) $\frac{20}{1.6} \times 10^3 \text{ ms}^{-1}$ 2) $10 \times 10^3 \text{ ms}^{-1}$ 3) $20 \times 10^3 \text{ ms}^{-1}$ 4) $40 \times 10^3 \text{ ms}^{-1}$

Sol:
$$V = \frac{E}{B} = \frac{20 \times 10^3}{2} = 10 \times 10^3 \text{ m/s}$$

Physics - Question 112

Original paper page 31; original question and option regions

112. A material of 0.25 cm^2 cross sectional area is placed in a magnetic field of strength (H) 1000 Am^{-1} . Then the magnetic flux produced is (Susceptibility of material is 313) (Permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$)

1) 8.33×10^{-8} weber 2) 1.84×10^{-6} weber 3) 9.87×10^{-6} weber 4) 3.16×10^{-6} weber

Sol: $\phi = \vec{B} \cdot \vec{A} = BA = \mu HA = \mu_0 (1 + \chi) HA = 4\pi \times 10^{-7} [314] \times 10^3 \times 25 \times 10^{-6}$
 $= 9 \times 10^{-6} \text{ weber}$

Physics - Question 113

Original paper page 31; original question and option regions

113. The magnitude of the induced emf in a coil of inductance 30 mH in which the current changes from 6A to 2A in 2 sec. is

1) 0.06V

2) 0.6V

3) 1.06V

4) 6V

Sol:
$$e = -L \frac{di}{dt} = -30 \times 10^{-3} \left(\frac{4}{2} \right) = 0.06A$$

Physics - Question 114

Original paper page 31; original question and option regions

114. In an AC circuit V and I are given below, then find the power dissipated in the circuit

$$V = 50 \sin (50t) \text{ V}, I = 50 \sin \left(50t + \frac{\pi}{3} \right) \text{ mA}$$

1) 0.625 W

2) 1.25 W

3) 2.50 W

4) 5.0 W

Physics - Question 114 (continued)

Original paper page 32; original question and option regions

Sol:
$$\text{power} = V_{\text{rms}} i_{\text{rms}} \cos \phi = \frac{50}{\sqrt{2}} \times \frac{50}{\sqrt{2}} \cos \frac{\pi}{3} = \frac{2500}{2} \times \frac{1}{2} = 625 \times 10^{-3} \text{ watt}$$

Physics - Question 115

Original paper page 32; original question and option regions

115. Light with an energy flux of 9 W cm^{-2} falls on a non-reflecting surface at normal incidence. If the surface has an area of 20 cm^2 . The total momentum delivered for complete absorption in one hour is

- 1) $2.16 \times 10^{-4} \text{ kgms}^{-1}$ 2) $1.16 \times 10^{-3} \text{ kgms}^{-1}$ 3) $2.16 \times 10^{-3} \text{ kgms}^{-1}$ 4) $3.16 \times 10^{-4} \text{ kgms}^{-1}$

Sol: $I = \frac{E}{At}$

$$9 = \frac{E}{20 \times 3600} \Rightarrow E = 9 \times 20 \times 3600$$

$$p = \frac{E}{C} = 2.16 \times 10^{-3} \text{ kgm/s}$$

Physics - Question 116

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116. The ratio of the deBroglie wave lengths for the electron and proton moving with the same velocity is (m_p -mass of proton, m_e -mass of electron)

1) $m_p : m_e$

2) $m_p^2 : m_e^2$

3) $m_e : m_p$

4) $m_e^2 : m_p^2$

Sol:
$$\frac{\lambda_e}{\lambda_p} = \frac{h / m_e V}{h / m_p V} = \frac{m_p}{m_e}$$

Physics - Question 117

Original paper page 32; original question and option regions

117. The ratio of longest wavelength lines in the Balmer and Paschen series of hydrogen spectrum is

1) $\frac{5}{16}$

2) $\frac{7}{20}$

3) $\frac{7}{144}$

4) $\frac{5}{27}$

Sol: $\lambda_{\text{Balmer}} \propto \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$

$$\lambda_{\text{Paschen}} \propto \left[\frac{1}{3^2} - \frac{1}{4^2} \right]$$

$$\frac{\lambda_1}{\lambda_2} = \frac{7}{20}$$

Physics - Question 118

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118. In the following nuclear reaction 'x' stands for $n \rightarrow p + e^- + x$

1) α -particle

2) positron

3) neutrino

4) Antineutrino

Sol: $n \rightarrow p + e^- + \bar{\nu}$

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Physics - Question 118 (continued)

Original paper page 33; original question and option regions

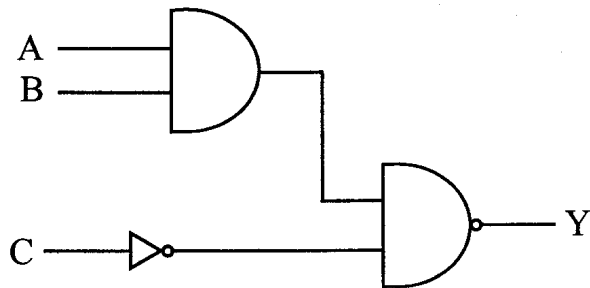
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Physics - Question 119

Original paper page 33; original question and option regions

119. In the following circuit the output Y becomes zero for the input combinations



1) $A = 1, B = 0, C = 0$

2) $A = 0, B = 1, C = 1$

3) $A = 0, B = 0, C = 0$

4) $A = 1, B = 1, C = 0$

Sol: $\overline{(AB)}.\overline{\overline{C}} = Y$

$$\overline{(1.1)}\overline{0} = \overline{(1.1)}.1 = 0$$

Physics - Question 120

Original paper page 33; original question and option regions

120. The maximum amplitude of an amplitude modulated wave is 16V, while the minimum amplitude is 4V. The modulation index is

1) 0.4

2) 0.5

3) 0.6

4) 4

Sol: $E_c + E_m = 16$

$$E_c - E_m = 4$$

$$2E_c = 20$$

$$E_c = 10$$

$$E_m = 6$$

$$\mu = \frac{E_m}{E_c} = \frac{6}{10} = 0.6$$

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Physics - Question 120 (continued)

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